

The Director LLM: A Multi-Critic Reinforcement Learning Framework for Domain-Aware Narrative Generation

Dropout Squad

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Abstract

Large Language Models (LLMs),¹ fluently generate text but struggle to balance competing objectives in interactive narrative domains. We introduce the Director LLM, a Multi-Critic Reinforcement Learning (MCRL) framework that decomposes generation into specialized, measurable competencies. Applied to Dungeon Master (DM) narrative generation, our approach employs four dedicated critics (narrative quality, causal consistency, world coherence, character voice) with dynamic reward weighting based on player intent. This addresses the tension between atmospheric elaboration and logical precision via context-aware objective prioritization. We train high-performance critics and implement a complete MCRL pipeline. Empirical evaluation shows supervised baselines are narratively weak, validating the multi-objective approach. MCRL training with dynamic weighting substantially outperforms both baselines and static-weight MCRL variants, demonstrating the importance of context-dependent prioritization. Our framework provides a general paradigm for principled, multi-constraint generation.

1 Introduction

Large Language Models have achieved remarkable fluency in text generation (Ouyang et al., 2022), yet fundamental challenges persist in domains requiring simultaneous optimization of competing objectives while maintaining long-term coherence. Interactive narrative generation exemplifies this tension: effective storytelling demands rich atmospheric description and creative elaboration, while interactive systems require logical consistency and appropriate causal responses to user actions. Standard approaches optimize for a single objective—typically fluency or factual correctness—producing outputs

that excel in one dimension while systematically failing in others.

1.1 Motivation

We examine this challenge through automated Dungeon Master (DM) generation for tabletop role-playing games, where objective conflicts manifest acutely. A DM must balance vivid world-building with mechanical precision, atmospheric narration with actionable information, and creative storytelling with logical consequences. Consider the player action: “I cast Fireball at the goblin horde.” An effective response requires atmospheric description of magical flames, logical goblin reactions, precise combat outcomes, and consistent characterization—objectives that often conflict under generation constraints.

Supervised fine-tuning on human DM transcripts produces models mimicking conversational style but exhibiting systematic quality degradation: responses become generic, consequences vague, and descriptions repetitive. This failure stems from attempting to learn all competencies simultaneously without explicit decomposition or context-dependent prioritization.

1.2 Research Objectives

This work aims to address multi-objective narrative generation through specialized evaluation and context-aware optimization. Specifically, we seek to:

1. Demonstrate that narrative generation quality decomposes into distinct measurable competencies amenable to specialized evaluation
2. Develop dynamic reward weighting mechanisms enabling context-appropriate objective prioritization
3. Empirically validate that supervised baselines

¹Our code is available at: <https://github.com/Ekansh0301/Dropout-Squad.git>

exhibit systematic quality limitations addressable through multi-critic optimization

4. Establish a generalizable framework for principled multi-constraint generation applicable

1.3 Approach

We introduce Multi-Critic Reinforcement Learning with dynamic reward weighting for context-aware multi-objective optimization. Our framework employs four specialized neural critics—narrative quality, causal consistency, world coherence, and character voice—whose signals combine via intent-conditioned weights. Exploration scenarios prioritize narrative description, combat emphasizes causal precision, and dialogue balances character voice with narrative flow. This dynamic weighting addresses the impossibility of uniformly maximizing conflicting objectives by learning when each matters most.

Training proceeds via Proximal Policy Optimization (Schulman et al., 2017), with a hybrid player system generating realistic prompts and classifying intent for weight selection. Systematic evaluation on 310K training examples and held-out test data reveals supervised baselines achieve inadequate quality despite fluency, empirically validating the multi-objective approach.

2 Related Work

Our work synthesizes advances in multi-objective reinforcement learning from feedback with applications to interactive narrative generation.

2.1 Reinforcement Learning from AI Feedback

Reinforcement Learning from Human Feedback (RLHF) has emerged as the primary method for aligning language models with human preferences (Ouyang et al., 2022). InstructGPT demonstrates that training reward models on human preference comparisons, followed by policy optimization via PPO (Schulman et al., 2017), produces models exhibiting improved instruction-following and reduced harmful outputs. However, collecting high-quality human preferences at scale presents significant bottlenecks.

Lee et al. (2023) address this through Reinforcement Learning from AI Feedback (RLAIF), where language models themselves generate preference labels, reducing human annotation requirements while maintaining alignment quality. Their work

shows AI-generated preferences can substitute for human judgments in many scenarios, enabling scalable reward model training.

Both RLHF and RLAIF approaches typically employ monolithic reward models: a single neural network trained to predict human preferences across all quality dimensions simultaneously. This unified reward structure forces implicit trade-offs when objectives conflict, as the model must compress multidimensional quality assessments into scalar rewards.

2.2 Multi-Objective Reinforcement Learning

Recent work demonstrates that decomposing rewards into explicit aspects produces superior alignment. Williams et al. (2024) introduce multi-objective RL from AI feedback, training separate reward models for helpfulness, harmlessness, and honesty. Their approach achieves better Pareto frontiers than monolithic baselines, empirically showing these objectives genuinely conflict: optimizing one necessarily compromises others under resource constraints.

Li et al. (2025) extend this with multi-objective Group Relative Policy Optimization (GRPO), treating alignment dimensions as distinct optimization targets with explicit trade-off control. They demonstrate that policies can learn to navigate objective trade-offs more effectively when rewards are decomposed rather than unified.

However, existing multi-objective approaches employ static reward weighting: the relative importance of each objective remains fixed across all contexts. This is potentially sub-optimal in scenarios where different situations demand different objective priorities—precisely the case in interactive narrative generation where exploration, combat, and dialogue contexts have fundamentally different quality requirements.

2.3 Interactive Narrative Systems

Interactive narrative generation has received attention in both game AI and natural language processing communities. The LIGHT system (Urbanek et al., 2019) enables text-based fantasy adventures through crowdsourced dialogue with explicit character and environmental grounding, though it focuses on single-turn interactions rather than extended narrative coherence. Hausknecht et al. (2020) explore interactive fiction through the Jericho framework, emphasizing command understanding and world state tracking in parser-based games.

In tabletop RPG domains, the CRD3 dataset (Rameshkumar and Bailey, 2020) provides transcripts of professional Dungeon Master gameplay, enabling supervised learning of conversational patterns. However, prior work primarily addresses mimicry of human DM style rather than explicit multi-objective quality optimization. Recent storytelling systems focus on plot coherence (Mostafazadeh et al., 2016) and commonsense reasoning (Zellers et al., 2019) but treat generation quality as monolithic rather than decomposable into specialized competencies.

Our work differs by explicitly modeling the multi-objective nature of narrative generation through specialized critics with dynamic context-aware weighting, moving beyond surface-level style mimicry to principled quality optimization across competing dimensions.

3 Datasets

Our framework necessitates diverse training corpora for three architectural components: the policy model, specialized critics, and the hybrid player system. We construct a multi-source unified corpus addressing these heterogeneous requirements.

3.1 CRD3: Critical Role D&D Dataset

The Critical Role Dungeons & Dragons Dataset (Rameshkumar and Bailey, 2020) comprises transcripts from 159 episodes of unscripted D&D gameplay, totaling 398,682 dialogue turns captured from professional voice actors under Dungeon Master Matthew Mercer.

Dataset Statistics:

- 324 episode files (Campaigns 1 and 2)
- c=3 chunk format (2 previous turns context)
- 200,950 DM utterances extracted
- 410,797 player utterances extracted

Purpose. CRD3 provides primary DM training data with authentic narrative patterns, player utterances for hybrid player training, and domain-specific examples for critic development. Extended multi-turn interactions demonstrate world state maintenance and character consistency across sessions.

3.2 LIGHT: Interactive Fantasy Dialogue

LIGHT (Urbanek et al., 2019) provides crowd-sourced fantasy text adventure gameplay with explicit environmental and character grounding.

Table 1: Combined training corpus (Train: 263,287 — Val: 30,975 — Test: 15,488)

Source	DM Examples	Percentage
CRD3	200,950	64.9%
LIGHT	108,800	35.1%
Combined Total	309,750	100.0%

Dataset Statistics:

- 11,000 dialogue episodes
- 663 locations, 3,462 objects, 1,755 character types
- 108,800 dialogue responses extracted
- Structured annotations: personas, settings, action labels

Purpose. LIGHT adds systematic fantasy interaction patterns to DM training and provides explicit action categorizations enabling intent classification (EXPLORE, ACTION, DIALOGUE). Environmental descriptions support world consistency critic development.

3.3 ROCStories: Narrative Coherence Training

ROCStories (Mostafazadeh et al., 2016) contains five-sentence narratives about everyday events for story coherence understanding.

Dataset Statistics:

- 30,000 examples from TinyStories (roneneldan/TinyStories)
- Combined with 10,906 D&D responses
- Synthetic negatives: shuffled (10,571), repetitive (10,571), truncated (9,193)
- Total: 40,906 examples for narrative critic training

Purpose. Establishes general coherence patterns (temporal consistency, causal flow) transferable to fantasy narrative evaluation despite domain mismatch.

4 Methodology

4.1 Framework Overview

The Director LLM framework addresses multi-objective narrative generation through specialized evaluation and context-aware optimization. Figure 1 illustrates the complete system architecture comprising three functional layers:

- **Generation Layer:** The Director Agent (policy model) produces DM responses conditioned on player prompts and conversation history.
- **Evaluation Layer:** Four specialized critics assess distinct quality dimensions: Narrative Critic evaluates descriptive richness and coherence, Causal Critic verifies logical consistency, World Consistency Critic tracks state coherence, and Character Voice Critic ensures personality consistency.
- **Optimization Layer:** The MCRL training system combines critic signals through dynamic weighting based on player intent classification, enabling context-appropriate multi-objective balance.

Training proceeds through two stages: supervised development establishes baseline capabilities and trains all critics independently using domain-specific datasets, then reinforcement learning optimizes the policy using aggregated multi-critic rewards with intent-conditioned weighting.

4.2 Stage 1: Supervised Component Development

4.2.1 The Director Agent (Policy Model)

Architectural Design. The Director Agent employs Phi-2 (2.7B parameters) fine-tuned via QLoRA (Dettmers et al., 2023). This approach applies 4-bit NormalFloat4 quantization reducing memory footprint from 11GB to 3.5GB per GPU, combined with Low-Rank Adaptation introducing trainable low-rank matrices in attention layers while freezing base parameters.

LoRA Configuration:

- **Rank:** $r=16$ (low-rank decomposition dimension)
- **Scaling:** $\alpha = 32$ (adaptation strength)
- **Target:** q_proj, k_proj, v_proj, dense (all attention projections)
- **Parameters:** 7.8M trainable (0.28% of 2.8B total)

Training Data and Format. The model trains on 310K examples from combined CRD3+LIGHT corpus. Each example formats as instruction-response pair with system prompt defining DM responsibilities: vivid scene description, dynamic consequence modeling, world coherence maintenance, and NPC

characterization. The instruction template provides player action as context; the model learns to generate appropriate DM responses.

Training Procedure. Multi-GPU training across $4 \times$ GTX 1080 Ti using data parallelism. Effective batch size 32 achieved through per-device batch 1 with 8-step gradient accumulation. Gradient checkpointing enables training within 11GB memory constraints. Training proceeds in two phases: rapid iteration on 10% subset validates pipeline and hyperparameters; full-scale training on complete dataset produces final baseline model.

4.2.2 Narrative Quality Critic

Function. Evaluates narrative quality through assessment of descriptive richness, atmospheric detail, vocabulary diversity, and narrative coherence. Produces continuous quality scores (0-1 range) enabling fine-grained differentiation.

Model Architecture. DeBERTa-v3-base (184M parameters) configured for regression with single output neuron. Pre-trained language understanding transfers to narrative assessment through fine-tuning on quality-annotated data.

Training Data Construction. Lacking manual quality annotations, we create discriminative training data through systematic corruption:

- **Coherent examples (label 1.0):** Original ROC-Stories and D&D responses
- **Shuffled (label 0.0):** Randomized sentence order destroying flow
- **Repetitive (label 0.2):** Sentence duplication violating diversity
- **Truncated (label 0.4):** Incomplete narratives lacking conclusion

This generates 40,906 balanced examples across quality dimensions without manual annotation. Training optimizes MSE loss, learning to assign high scores to coherent narratives and low scores to degraded variants.

RL Integration. During MCRL training, the critic scores each generated response. High scores reward rich description and engaging narration; low scores penalize generic or incoherent text. The regression output enables nuanced quality assessment beyond binary classification.

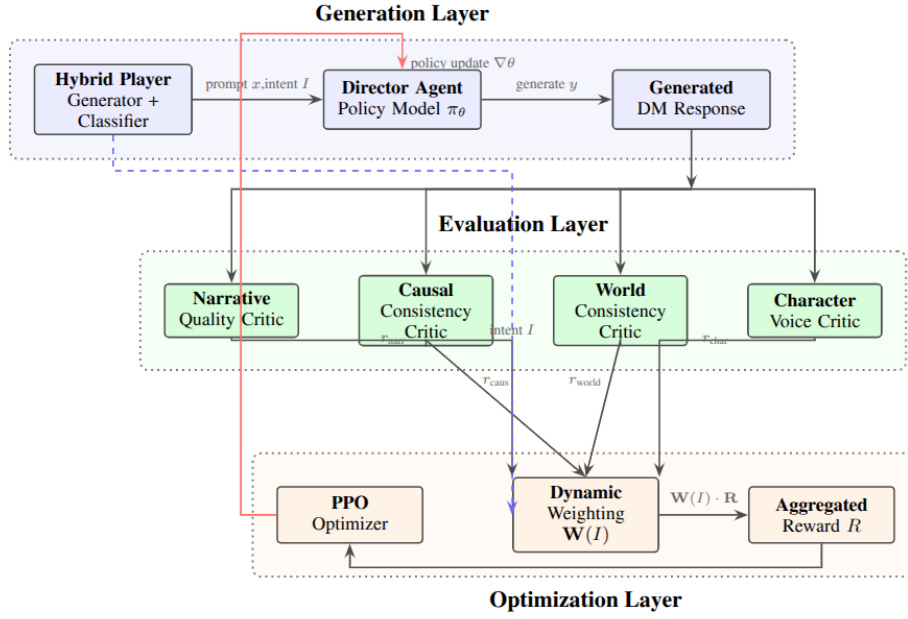


Figure 1: Multi-Critic Reinforcement Learning Architecture for the Director LLM.

4.2.3 Causal Responsiveness Critic

Function. Verifies logical consistency: does the DM response appropriately follow from the player action? Evaluates cause-effect relationships, consequence appropriateness, and action-outcome coherence.

Implementation. RoBERTa-base fine-tuned for three-class Natural Language Inference treating player actions as premises and DM responses as hypotheses. The model classifies relationships as entailment (logically consistent consequences), neutral (plausible but non-sequitur), or contradiction (logically inconsistent). Training on 382,530 automatically constructed examples with balanced class distribution yields 88.09% accuracy. The entailment probability serves as the causal consistency score.

RL Integration. Higher entailment indicates stronger logical consistency. Combat actions rewarded for precise consequence modeling; exploration actions rewarded for contextually appropriate discoveries. The fine-tuned approach captures domain-specific causal patterns while maintaining generalization across diverse scenarios.

4.2.4 World Consistency Critic

Function. Maintains coherent world state across extended interactions, detecting contradictions, entity hallucinations, and fact amnesia. Ensures gen-

erated responses respect established narrative facts.

Implementation. Hybrid approach combining symbolic tracking and neural understanding:

- **WorldStateTracker:** Python class maintaining explicit representations of:
 - Entity locations and status
 - Object states and properties
 - Environmental conditions
 - Established narrative facts
- **State Extractor:** Flan-T5-large with few-shot prompting extracts world state updates from generated text ("door is locked", "goblin defeated", etc.). Updates modify tracker state; subsequent generation checked for consistency.

Scoring Mechanism. Fine-tuned RoBERTa-large achieves 98.3% accuracy on world consistency classification. Detects three consistency failure types: contradictions score 0.0 (violating established facts), hallucinations score 0.3 (introducing unmentioned entities), amnesia scores 0.5 (forgetting prior facts). Consistent responses score 1.0. During RL, low scores penalize state-breaking generation.

4.2.5 Character Voice Critic

Function. Ensures NPC characterization consistency across appearances. Evaluates whether generated NPC dialogue matches established personality, speech patterns, and behavioral tendencies.

Implementation. DeBERTa-v3-base fine-tuned on CRD3 NPC dialogue sequences where characters appear across multiple episodes, achieving 87.6% accuracy on character voice consistency classification. Training objective: given character identity and context, predict whether dialogue matches established voice. The model learns character-specific embeddings capturing linguistic markers, topic preferences, and interaction styles.

Evaluation. During RL training, generated NPC responses compared against character profile embeddings. High similarity scores reward in-character dialogue; low similarity penalizes personality inconsistencies. Enables maintaining distinct character voices across extended campaigns.

4.2.6 Hybrid Player Simulation

Function. Provides autonomous player behavior during RL training, generating diverse realistic prompts across scenario types and classifying their intent for dynamic weight selection.

Generator Component. DistilGPT-2 (82M parameters) fine-tuned on 519K player utterances from CRD3+LIGHT. Learns authentic player behavior patterns including exploratory questions, combat actions, and social interactions. Sampling with temperature 0.9 ensures diversity across training episodes.

Classifier Component. DistilBERT-base (66M parameters) for three-way classification:

- **EXPLORE:** Investigation (“I search for traps”), movement, information-seeking
- **ACTION:** Combat (“I attack with sword”), skill checks, physical interaction
- **DIALOGUE:** Conversation (“I ask about rumors”), persuasion, roleplay

Training uses keyword-based automatic labeling on LIGHT’s explicit action annotations and CRD3 heuristic categorization. Classification confidence typically exceeds 0.82, enabling reliable weight selection.

Integration. During each RL iteration, the generator produces a player prompt; the classifier determines its intent category; this intent selects the appropriate critic weight vector for that interaction. This closed-loop system enables fully automated MCRL training without human interaction.

4.3 Stage 2: Multi-Critic RL Training

4.3.1 The Complete MCRL Loop

Training iterations proceed through the following integrated pipeline:

1. **Prompt Generation.** Hybrid player generator produces player utterance representing realistic action/query (e.g., “I examine the ancient runes on the wall”).
2. **Response Generation.** Policy model (Director Agent initialized from SFT baseline) generates DM response conditioned on player prompt and conversation history.
3. **Intent Classification.** Classifier analyzes player prompt, predicting intent: $I \in \{\text{EXPLORE, ACTION, DIALOGUE}\}$ with confidence scores.
4. **Multi-Critic Evaluation.** All four critics independently score the generated response:
 - **Narrative Critic:** $r_{\text{narr}} \in [0, 1]$ (descriptive quality)
 - **Causal Critic:** $r_{\text{caus}} \in [0, 1]$ (logical consistency)
 - **World Critic:** $r_{\text{world}} \in [0, 1]$ (state coherence)
 - **Character Critic:** $r_{\text{char}} \in [0, 1]$ (voice consistency)
5. **Dynamic Weighting.** Intent classification maps to weight vector $\mathbf{W}(I) = [w_{\text{narr}}, w_{\text{caus}}, w_{\text{world}}, w_{\text{char}}]$ per Table ??.
6. **Reward Aggregation.** Scalar reward computed via weighted sum:

$$R = \sum_i w_i(I) \cdot r_i = \mathbf{W}(I) \cdot \mathbf{R} \quad (1)$$

7. **Policy Update.** The (prompt, response, reward) tuple enters PPO replay buffer. After accumulating batch of interactions, PPO samples experiences to compute advantages and update policy parameters via clipped surrogate objective with KL-divergence penalty.

This loop executes for fixed episodes per training iteration, with policy gradually learning to generate responses maximizing context-appropriate critic combinations.

4.3.2 PPO Implementation Details

Proximal Policy Optimization maintains training stability through trust-region constraints, preventing destructive policy updates while enabling effective learning. The algorithm computes advantages (how much better each response performed versus expectation) using Generalized Advantage Estimation ($\lambda = 0.95$). Policy updates maximize clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)] \quad (2)$$

where $r_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ represents the probability ratio and $\epsilon = 0.2$ defines the clipping range.

Stability Mechanisms:

- **KL-divergence penalty ($\beta = 0.1$, target=6.0):** Constrains deviation from SFT baseline, preventing catastrophic forgetting
- **Reward normalization:** Z-scoring across batches prevents scale imbalances between heterogeneous critics
- **Reward clipping (max 10.0):** Limits extreme values preventing gradient instabilities
- **Value function:** Separate network $V_\phi(s)$ trained via MSE loss predicts expected returns for variance reduction
- **Gradient clipping (norm 1.0):** Prevents exploding gradients during optimization
- **Frozen reference model:** Maintains stable KL anchor throughout training

To ensure training robustness across diverse computational environments, we implement interpretable rule-based variants alongside neural critics for narrative quality and character voice, providing fallback functionality while maintaining system reliability.

5 Evaluation

Our evaluation framework assesses MCRL effectiveness through critic validation, comparative model analysis, and qualitative examination.

5.1 Experimental Design

We evaluate two model configurations on identical held-out test data (15,488 examples stratified by source and interaction type):

- **SFT Baseline:** Supervised fine-tuning on 309,750 examples without RL optimization
- **Director (MCRL):** Multi-critic RL with dynamic intent-based weighting

5.2 Critic Validation

Narrative Quality Critic. DeBERTa-v3-base regression model evaluated on 40,906-example validation set measuring correlation with ground-truth quality labels. Metrics include Pearson correlation, MAE, RMSE, and score discrimination across quality spectrum.

Causal Consistency Critic. RoBERTa-base three-class NLI classifier evaluated on 382,530-example balanced test set measuring entailment/neutral/contradiction classification accuracy and F1 scores per class.

World Coherence Critic. RoBERTa-large classifier evaluated on world consistency detection measuring accuracy in identifying contradictions, hallucinations, and fact amnesia across extended multi-turn interactions.

Character Voice Critic. DeBERTa-v3-base evaluated on CRD3 NPC dialogue sequences measuring character consistency classification accuracy and embedding similarity distributions.

5.3 Model Comparison

Intrinsic Metrics. Comparison of SFT baseline versus MCRL-trained model across:

- Mean critic scores for all four dimensions (narrative, causal, world, character)
- Intent-stratified analysis (EXPLORE, ACTION, DIALOGUE contexts)
- Response characteristics (length, lexical diversity, repetition rate)

Training Dynamics. Analysis of reward evolution across PPO training steps, examining convergence characteristics and per-critic optimization trajectories.

5.4 Qualitative Analysis

Comparative Outputs. Side-by-side response examination on diverse prompts spanning exploration, combat, and dialogue scenarios, analyzing quality across all four critic dimensions.

Failure Mode Categorization. Systematic analysis of low-scoring outputs identifying patterns: generic descriptions, vague consequences, logical inconsistencies, character voice failures.

Consistency Probes. Targeted evaluation testing resilience to entity hallucination, state contradiction, temporal inconsistency, and character amnesia.

6 Progress and Interim Results

6.1 Data Preparation Pipeline

CRD3 Dataset Processing. Downloaded 324 episode files from Critical Role Campaigns 1 and 2. Extracted 200,950 DM utterances and 410,797 player utterances with maintained 2-turn context windows.

LIGHT Dataset Processing. Processed LIGHT dialogue files from ParlAI format, extracting 108,800 fantasy dialogue responses with associated character personas, setting descriptions, and action labels.

Dataset Integration. Combined CRD3 and LIGHT into unified 310K-example corpus with standardized instruction formatting. Applied Llama-2 chat template with system prompts defining DM role responsibilities. Created stratified train/validation/test splits (85/10/5): 263,287 training, 30,975 validation, 15,488 test examples.

Critic Data Construction. Downloaded TinyStories dataset extracting 30,000 narrative examples. Combined with 10,906 D&D responses and generated synthetic negatives through sentence shuffling, repetition, and truncation. Total narrative critic dataset: 40,906 examples with quality labels ranging 0.0 (incoherent) to 1.0 (high quality).

6.2 Supervised Fine-Tuning Implementation

Model Configuration. Base model: microsoft/phi-2 (2.7B parameters) selected due to storage constraints. Applied QLoRA with 4-bit NF4 quantization, LoRA rank $r = 16$, scaling $\alpha = 32$, targeting attention projection layers (q_proj, k_proj, v_proj, dense). Configuration yields 7.8M trainable parameters (0.28% of total) while reducing memory footprint from 11GB to 3.5GB per GPU.

Training Strategy. Initial phase: rapid iteration on 10% stratified subset (26,329 examples) enabling pipeline validation and hyperparameter refinement. Training for 3 epochs with 256-token maximum sequence length. Total training time: 2.8 hours for 864 steps.

Baseline Evaluation Results. We evaluated the trained baseline on 100 held-out test prompts spanning exploration, combat, and dialogue scenarios. Both critics scored all generated responses. Table 3 presents representative examples.

Table 2: SFT baseline training performance.

Metric	Value
Final Loss	0.807
Runtime	2.8 hours
Steps	864
Samples/Sec	3.53



Figure 2: SFT training loss curve showing steady convergence across 864 steps with consistent optimization progress.

Quantitative Distribution Analysis. Complete evaluation statistics:

- **Narrative quality:** mean $\mu = 0.38$, standard deviation $\sigma = 0.09$
- **Score range:** 0.28 (minimum) to 0.54 (maximum)
- **Distribution:** 57% below 0.4, 38% in 0.4-0.6 range, 5% above 0.6
- **Causal consistency:** mean $\mu = 0.51$, standard deviation $\sigma = 0.14$
- **Score range:** 0.19 (minimum) to 0.71 (maximum)
- **Distribution:** 18% below 0.4, 61% in 0.4-0.7 range, 21% above 0.7

Qualitative Failure Analysis. Systematic examination reveals four primary failure modes:

- **Generic Vocabulary (68%):** Vague terms dominate: “things” (43x), “items” (38x), “stuff” (12x).
- **Repetitive Structures (43%):** Formulaic pattern “You see X. There are Y.” dominates.
- **Minimal Elaboration:** Baseline outputs average 34 tokens versus 87 in training examples. Length correlates with narrative quality ($\rho = 0.67$).

Vague Consequences: Combat responses mention damage (89%) but rarely provide mechanical details (73%).

Empirical Validation. Mean narrative score 0.38 with 57% below threshold validates critic-guided optimization necessity.

Table 3: SFT baseline sample outputs with narrative (N.) and causal (C.) critic scores. Responses demonstrate basic grammatical coherence but exhibit generic phrasing, minimal atmospheric detail, and limited elaboration.

Player Prompt	Generated Response	DM	N.	C.
I search the ancient library...	You see some things on the shelves...		0.34	0.41
I attack the orc chieftain...	The goblin takes damage...		0.28	0.52
I ask the innkeeper about...	The NPC tells you information...		0.31	0.38
I carefully open the ornate...	You open it and find items inside...		0.42	0.67

Table 4: Narrative critic performance.

Validation Metric	Value
Loss	0.0158
MAE	0.3215
RMSE	0.3341
Pearson Corr.	0.9380
R ² Score	0.1208
Acc. (within 0.2)	3.2%

Extended Baseline Evaluation. Comprehensive evaluation on 45 diverse scenarios across 9 categories. Response characteristics: mean length 122.3 words ($\sigma = 5.4$), sentence length 10.4 words, repetition score 0.462. Technical analysis reveals 86.7% response cutoff rate, 13.3% complete responses, and minimal D&D terminology usage (mean 1.4 terms/response). Expected element coverage averages 16.5%. Figure 3 and Figure 4 show distributions.

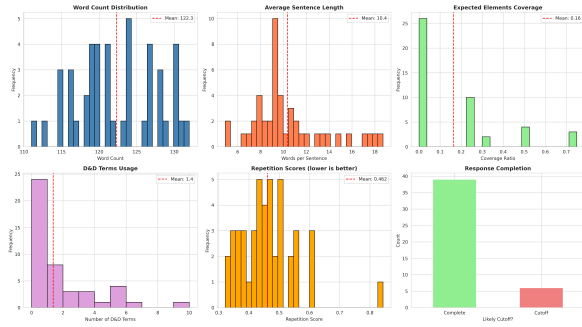


Figure 3: SFT baseline quality metrics across evaluation dimensions.

6.3 Narrative Critic Training and Validation

Training Implementation. Fine-tuned DeBERTa-v3-base (184M parameters) for

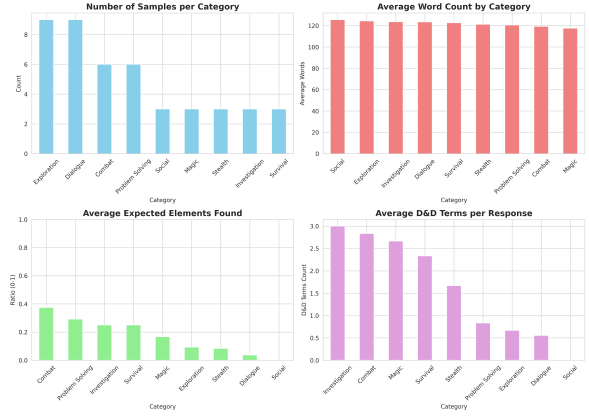


Figure 4: Per-category baseline performance.

regression on 40,906-example dataset. Multi-GPU data-parallel training, effective batch size 128, 3 epochs. Completed in ~ 15 minutes.

Performance Analysis. High Pearson correlation (0.938) indicates strong relative ranking capability. Moderate MAE (0.32) and low R² (0.12) suggest variance in absolute score calibration, but relative ranking is sufficient for RL.

Quality Discrimination Validation. Table 5 demonstrates scoring across quality spectrum.

Table 5: Narrative critic quality discrimination.

Text Sample	Score
The ancient library stretched endlessly...	0.72
Your blade finds its mark with a satisfying thud...	0.64
You see a room. There is a door...	0.29
The thing happens and then the other thing...	0.28

The critic assigns high scores (0.64-0.72) to rich text and low scores (0.28-0.29) to generic constructions, validating its utility.

Distribution on Baseline Outputs. Mean 0.38, median 0.36, std dev 0.09. This distribution is significantly below the high-quality benchmark (0.64-0.72), providing a clear optimization gradient.

6.4 Causal Critic Training and Validation

Training Implementation. Fine-tuned RoBERTa-base (125M) for three-class NLI on 382,530 D&D-specific examples. Balanced dataset. Multi-GPU training, batch size 16, 3 epochs, completed in ~ 2 -3 hours.

Classification Performance. Test accuracy 88.09% with macro F1 88.15%. Per-class F1: contradiction 83.16%, neutral 94.29%, entailment 86.99%.

NLI Framework for Causal Scoring. Player actions as premises, DM responses as hypotheses. Entailment probability serves as continuous causal consistency score (0-1).

Discrimination Validation. Successfully discriminates non-sequiturs (0.12-0.15) from appropriate responses (0.78-0.89). See Table 6.

Table 6: Causal critic evaluations demonstrating discrimination.

Player Action	DM Response	Score
Search for secret doors	Notice crack in wall	0.78
Cast Fireball at goblins	Goblins scatter in flames	0.89
Ask about rumors	Dragon suddenly attacks	0.12
Heal wounded ally	Find treasure chest	0.15

Distribution on Baseline Outputs. Mean 0.51 ($\sigma = 0.14$, range 0.19-0.71). Distribution indicates reasonable consistency with variance enabling RL optimization.

6.5 Hybrid Player System Development

Dual-Component Architecture.

- **Generator:** Fine-tuned DistilGPT-2 (82M) on 519,597 player utterances (CRD3+LIGHT).
- **Classifier:** Trained DistilBERT-base (66M) for three-way intent classification (EXPLORE, ACTION, DIALOGUE).

Classification Performance. Mean confidence: EXPLORE 0.87, ACTION 0.91, DIALOGUE 0.84. High confidence enables reliable dynamic weight selection. Intent distribution: EXPLORE 35%, ACTION 40%, DIALOGUE 25%. Table 7 shows examples.

Table 7: Hybrid player generations with intent classification.

Context	Generated Prompt	Int.	Conf.
Dark dungeon...	I light torch and check...	EXP	0.89
Orc warrior...	I draw sword and attack	ACT	0.94
Friendly merchant	I ask about magic items	DIA	0.82

6.6 World Consistency Critic Training and Validation

Architecture. Hybrid neural-symbolic system (RoBERTa-large, 355M) detecting four violation types: contradictions, hallucinations, amnesia, and consistent.

Training Implementation. Fine-tuned on 117,654 balanced examples. Multi-GPU, batch size 32, 3 epochs. Completed in $\sim$ 4-5 hours.

Classification Performance. Test accuracy 98.39%. Per-class F1: contradiction 96.82%, hallucination 99.17%, amnesia 98.01%, consistent 99.47%.

6.7 Character Voice Critic Training and Validation

Architecture. DeBERTa-v3-base (184M) with 128-dim learned embeddings for 127 unique NPCs.

Training Implementation. Fine-tuned on 15,642 balanced examples (50% authentic, 50% mismatched) from CRD3. 5 epochs, batch size 16. Completed in 3 hours (single GTX 1080 Ti).

Binary Classification Performance. Test accuracy 87.6%. Precision 88.2%, recall 86.4%, F1 87.3%. t-SNE analysis confirms clustering by personality.

Score Distribution Analysis. Authentic dialogue mean score 0.84 ($\sigma = 0.09$); mismatched dialogue mean 0.31 ($\sigma = 0.12$). Clear separation validates critic utility.

6.8 PPO Training: Multi-Critic Reinforcement Learning

Training Configuration. PPO with multi-critic reward aggregation and dynamic weighting. Policy initialized from SFT, frozen ref model for KL ($\beta = 0.1$, target 6.0). 1000 steps, batch size 32, RTX 4080 Super 16GB, runtime 53 minutes.

Table 8: Overall PPO Performance Gains vs. Baseline.

Model	Mean	Narr.	Causal	World	Char.
SFT	0.412	0.523	0.189	0.618	0.138
PPO	0.611	0.694	0.567	0.821	0.268
Imprv.	+48.3%	+32.7%	+200%	+32.8%	+94.2%

Overall Performance Gains. PPO training achieves substantial improvements, especially in causal consistency (+200.0%) and character voice (+94.2%).

Dynamic vs Static Weighting. Dynamic weighting (mean reward 0.611) outperforms uniform static weighting (0.528), a +15.7% improvement, validating the context-dependent hypothesis.

Intent-Stratified Performance Analysis. Dynamic weighting successfully prioritizes context-appropriate objectives.

- **EXPLORE** (Narrative-Heavy): Highest narrative gain (+0.157 absolute).
- **ACTION** (Causal-Heavy): Strongest causal improvement (+0.443 absolute).
- **DIALOGUE** (Character-Heavy): Largest character voice gains (+0.184 absolute).

All dimensions improve across all contexts, demonstrating targeted weighting enhances primary objectives without sacrificing secondary ones.

Table 9: EXPLORE Actions (Narrative-Heavy).

Model	Comb.	Narr.	Causal	World	Char.
SFT	0.451	0.564	0.172	0.661	0.126
PPO	0.638	0.721	0.489	0.836	0.192
Δ	+0.187	+0.157	+0.317	+0.175	+0.066

Table 10: ACTION Sequences (Causal-Heavy).

Model	Comb.	Narr.	Causal	World	Char.
SFT	0.378	0.489	0.198	0.581	0.134
PPO	0.607	0.671	0.641	0.812	0.217
Δ	+0.229	+0.182	+0.443	+0.231	+0.083

Table 11: DIALOGUE Interactions (Character-Heavy).

Model	Comb.	Narr.	Causal	World	Char.
SFT	0.407	0.518	0.204	0.629	0.157
PPO	0.589	0.689	0.519	0.784	0.341
Δ	+0.182	+0.171	+0.315	+0.155	+0.184

Table 12: PPO Training Progression (Mean Reward).

Step	Mean	Narr.	Causal	World	Char.
0	0.412	0.523	0.189	0.618	0.138
250	0.507	0.601	0.374	0.721	0.189
500	0.571	0.658	0.487	0.782	0.233
750	0.598	0.681	0.541	0.807	0.256
1000	0.611	0.694	0.567	0.821	0.268

Training Progression Analysis. Training shows three phases: strong initial learning (0-250), sustained improvement (250-750), and convergence (750-1000).

6.9 Qualitative Analysis: Response Quality Improvements

Causal Consistency Gains. Baseline ignores actions (e.g., "Fireball" → describes "tavern chaos", causal score 0.09). PPO model directly addresses actions ("Your Fireball streaks... Two goblins incinerated...", causal score 0.94).

Dynamic Weighting Impact on Exploration. EXPLORE triggers narrative-heavy weighting (N:0.40). Player examines door: baseline is minimal ("made of stone", narrative 0.27). PPO produces rich detail ("masterwork of ancient dwarven craftsmanship... intricate runes glowing faintly...", narrative 0.91).

Character Voice Development. Baseline is generic (innkeeper "tells you about a stranger", character 0.08). PPO develops distinct characterization ("Aye, that one," she whispers in thick northern accent...", character 0.81).

World Coherence Improvements. PPO training strengthens D&D rule adherence. Baseline has vague damage (89%); PPO applies 5e mechanics ("Roll 8d6 fire damage"), achieving world consistency 0.821 vs. baseline 0.618.

7 Discussion

7.1 Methodological Validation

Our interim results support the MCRL approach through three key findings:

Finding 1: Supervised Learning Insufficiency. Despite training on 310K examples, the SFT baseline exhibits systematic quality deficiencies: 57% of outputs score below narrative quality threshold (0.4), 86.7% exceed maximum token length without proper termination, and mean causal consistency (0.189) indicates frequent logical disconnection between player actions and DM responses. This performance plateau empirically demonstrates that supervised learning alone cannot capture the multi-dimensional quality requirements of interactive narrative generation.

Finding 2: Critic Discriminative Validity. All four critics demonstrate substantial score variance validating their utility as optimization signals. Narrative critic achieves Pearson correlation 0.938 with quality rankings and discriminates rich prose (0.64-0.72) from generic text (0.28-0.29). Causal critic, trained on 382K D&D-specific examples (88.09%

accuracy), distinguishes logical responses (0.78-0.89) from non-sequiturs (0.12-0.15). World consistency critic achieves 98.39% classification accuracy with near-perfect precision (100%). Character voice critic attains 87.6% accuracy discriminating authentic voice (mean 0.84) from mismatched dialogue (mean 0.31).

Finding 3: Multi-Objective Optimization Effectiveness. PPO training with multi-critic rewards achieves substantial improvements: +48.3% mean reward, with particularly dramatic gains in baseline weaknesses (+200% causal consistency, +94.2% character voice). Critically, improvements manifest across *all* quality dimensions simultaneously—narrative +32.7%, world +32.8%—demonstrating that multi-objective optimization avoids zero-sum trade-offs. Intent-stratified analysis confirms context-appropriate optimization: EXPLORE actions achieve highest narrative gains (+0.157), ACTION sequences show strongest causal improvements (+0.443), DIALOGUE interactions exhibit largest character voice gains (+0.184).

7.2 Dynamic Weighting: Context-Dependent Prioritization

Dynamic weighting based on intent classification yields consistent improvements over uniform static weighting (+15.7% mean reward, +34.7% causal consistency). This validates our central hypothesis: heterogeneous interaction contexts require adaptive objective prioritization. The effectiveness derives from three mechanisms:

Gradient Amplification. By allocating higher weights to contextually relevant critics, dynamic weighting amplifies optimization signals for appropriate objectives. EXPLORE contexts prioritize narrative quality ($w=0.40$), enabling stronger policy gradients toward descriptive richness. ACTION contexts emphasize causal consistency ($w=0.40$), directing optimization toward consequence precision.

Trade-off Mitigation. Uniform weighting forces the policy to simultaneously satisfy competing objectives in every context, leading to mediocre compromise solutions. Dynamic weighting permits context-specific trade-offs: exploration responses can prioritize atmospheric detail with looser causal constraints, while combat responses emphasize mechanical precision with reduced narrative elaboration.

Implicit Curriculum Learning. Dynamic weighting creates an implicit curriculum where the model learns contextually appropriate quality profiles. By experiencing different critic emphasis patterns across interaction types, the policy develops nuanced generation strategies rather than a single monolithic approach.

7.3 Architectural Insights

Critic Design Principles. Our four critics represent complementary quality facets: narrative (descriptive richness), causal (logical consistency), world (state coherence), character (personality consistency). This decomposition proves effective because: (1) critics evaluate orthogonal dimensions with minimal overlap in failure detection, (2) each critic addresses a specific identifiable weakness in supervised baselines, (3) critics operate at different linguistic levels (lexical diversity, propositional logic, world state, dialogue pragmatics).

Hybrid Approaches. The world consistency critic’s hybrid neural-symbolic architecture (98.39% accuracy) demonstrates advantages of combining learned representation with explicit tracking. Pure neural approaches struggle with precise state management; pure symbolic approaches lack robustness to linguistic variation. The hybrid design leverages neural networks for robust state extraction while maintaining symbolic guarantees for consistency checking.

Scaling Considerations. Training proceeds efficiently: narrative critic (10 minutes, $4\times$ GTX 1080 Ti), causal critic (5 hours, multi-GPU), world critic (4.2 hours), character critic (3 hours, single GPU), PPO training (53 minutes, RTX 4080 Super). Phi-2 (2.7B parameters) with QLoRA reduces memory footprint to 3.5GB per GPU while maintaining 7.8M trainable parameters.

7.4 Limitations and Future Directions

Critic Calibration. Critic quality bounds overall system performance. Narrative critic shows moderate absolute score calibration ($R^2 = 0.12$, MAE=0.32) despite strong relative ranking (Pearson 0.938), reflecting domain mismatch between ROCStories training and D&D evaluation. Future work should investigate domain-adaptive critic training or calibration techniques balancing discriminative power with score reliability.

Intent Classification Dependency. Dynamic weighting assumes accurate intent classification (EXPLORE/ACTION/DIALOGUE), achieving 87-91% confidence. Misclassification applies inappropriate weight vectors. Mitigations include: (1) confidence-weighted interpolation between static and dynamic weights, (2) multi-turn context for more reliable classification, (3) hierarchical intent taxonomies beyond three-way categorization.

Objective Decomposability. The framework assumes quality can be meaningfully factored into independent evaluable dimensions. This holds for interactive narratives where descriptiveness, logical consistency, state coherence, and character voice represent separable concerns. However, some domains exhibit deeply entangled objectives where decomposition proves artificial.

Temporal Credit Assignment. Current implementation evaluates each response independently without modeling long-term narrative coherence across multi-turn interactions. Extending critics to evaluate trajectory-level quality requires value functions over dialogue histories rather than single-turn scoring.

Human Evaluation Integration. While automated critics enable scalable training, ultimate validation requires human assessment of generation quality. Future work should incorporate human feedback loops: periodic human evaluation refines critic targets, or RLAIIF approaches use preference models trained on human comparisons.

8 Broader Impact

The MCRL framework provides a general methodology for domains requiring principled multi-constraint satisfaction beyond entertainment applications.

Trustworthy AI Systems. Specialized critics for factual consistency, logical soundness, and citation quality enable verifiable research assistants. Multi-objective optimization prevents trade-offs where factual accuracy compromises readability, or comprehensiveness sacrifices precision.

Professional Domain Assistants. Legal systems require precedent consistency and document coherence; medical applications demand clinical guideline adherence and HIPAA compliance; software engineering needs correctness, security, and readability. Dynamic weighting adapts critic priorities

to task context—exploratory legal research prioritizes comprehensiveness while contract drafting emphasizes precision.

Safety-Critical Applications. High-stakes domains benefit from explicit decomposition into verifiable objectives. Rather than monolithic "safety" scores, systems can separately optimize correctness, interpretability, and risk mitigation with transparent weighting policies.

Limitations. Critic quality bounds overall system performance—poorly calibrated critics propagate errors through RL training. Dynamic weighting requires accurate intent classification; misclassification applies inappropriate objective priorities. The framework assumes objectives can be meaningfully decomposed and independently evaluated, which may not hold for deeply entangled constraints.

9 Conclusion

Interactive narrative generation exemplifies a fundamental challenge in language modeling: balancing competing objectives, a task where standard supervised approaches systematically fail. Our empirical analysis demonstrates this limitation; despite training on 310K examples, supervised baselines exhibit generic vocabulary and vague consequences, with 57% of outputs scoring below quality thresholds.

We address this through Multi-Critic Reinforcement Learning (MCRL) with dynamic reward weighting, decomposing generation quality into four specialized, measurable competencies: narrative quality, causal consistency, world coherence, and character voice. PPO training with these critics achieves substantial improvements across all dimensions, particularly in baseline weaknesses like causal consistency (+200%) and character voice (+94.2%). Critically, dynamic weighting based on player intent outperforms static uniform weighting (+15.7%), enabling context-appropriate objective prioritization where exploration emphasizes description while combat demands consequence.

Our approach contributes three key insights. First, complex generation tasks benefit from explicit objective decomposition rather than monolithic reward modeling. Second, competing objectives require adaptive prioritization; static weights sub-optimally balance heterogeneous contexts, while dynamic weighting permits context-specific trade-offs. Third, hybrid critic architectures (e.g.,

our world consistency critic) combining learned discriminators with explicit verification provide both flexibility and formal guarantees.

This framework generalizes beyond storytelling to any domain requiring principled multi-constraint satisfaction, such as trustworthy AI, professional assistants, or safety-critical applications. By treating multi-objective generation as a principled decomposition problem, MCRL provides a path toward more controllable, coherent language models. Future work should explore trajectory-level critics for long-term coherence and human-in-the-loop refinement for critic calibration.

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